COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structure & Algorithms	**SECTION:**	CPE21S4
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Seatwork 8.1 **Using Sorting Algorithms 2**

1. PROCEDURE OUTPUT

Objectives: To create a program in C++ using search trees.

1. To be familiarized with the search tree algorithm.
2. To be able to create a program with search tree algorithm.

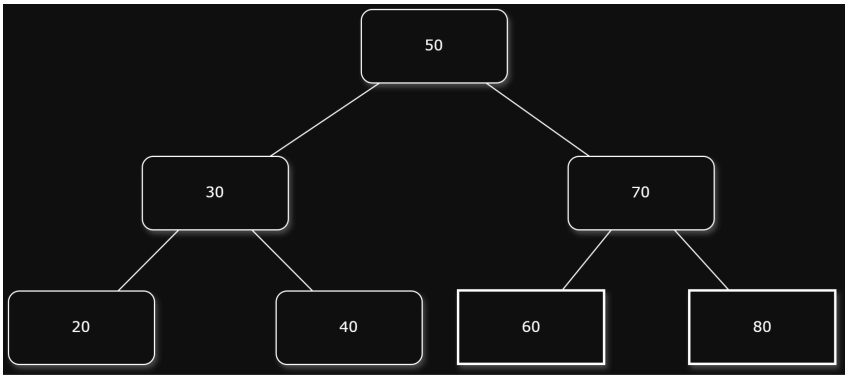
Answer the following questions:

1. What is a binary search tree?

- A binary search tree is a node-based data structure that is used to efficiently locate, store data in a sorted order to facilitate fast search, insertion, and deletion operations. As specified, the most common variant is the **Binary Search Tree (BST)**. The BST is a hierarchical structure where each node has at most two children. Each node contains a key along with references to its child nodes.

  Examples of a binary search tree could be storing the values 50, 30, 70, 20, 40, 60, and 80. In this case, 50 will be the root node, while all the lower values will be on the left side and higher values will be on the right side. In this way, searching becomes very easy because some nodes are simply ignored during the search process.

  Visual Aids:



2. Where can binary search tree be used?

- A Binary Search Tree can be used for faster lookups, traversal, insertion and deletion of data, as this data structure relies on the order of node insertion with arbitrary insertions that might cause some inefficiencies. Binary Search Trees can implement different abstract data types such as dynamic sets, lookup tables and priority queues; these are used in sorting algorithms for tree sorting.

3. What is a tree traversal.

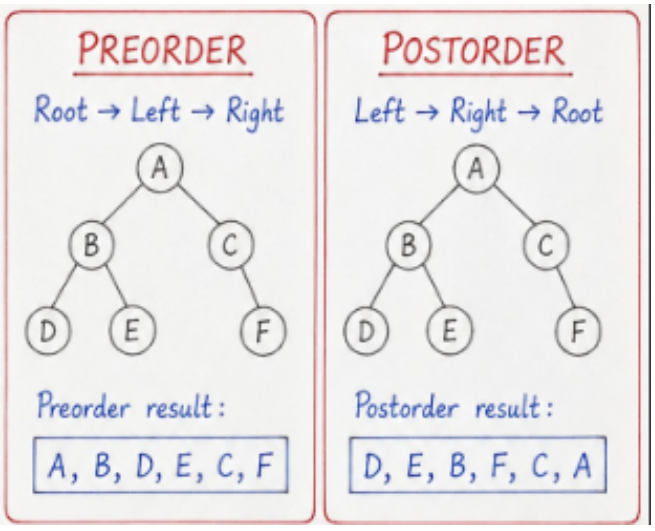
- Tree Traversal is the process of visiting each node in a tree structure exactly once in a specific, structured order. compared to linear data structures (arrays or linked lists), which have only one clear way to read through them, trees can be traversed in a variety of ways depending on the order in which you visit the root node and its relative to its children.

4. Differentiate a post-order and pre-order traversal give examples.

- The difference between pre-order and post-order traversal is about the order of traversing the structure in which the root, which is the parent node, is visited relative to its child nodes. Specifically;

    - **Pre-order traversal**: Visit the root node (parent node) **before** its children (child node).
    - **Post-order traversal**: Visit the root node (parent node) **after** visiting all of its children (child nodes).

    Examples:



5. What is a parse tree and its use?

- A parse tree is a hierarchical tree-like diagram that represents how a string or program is derived. It's the main blueprint for the binary search tree, where it consists of a root node, branch nodes which is also known as the child nodes, and the leaf nodes which are the bottom nodes that end of the parse tree or BST. Parse trees can be used to illustrate hierarchical relationships. In terms of DSA, it relates it to an input sentence or code expression. It also can be a visualizer / structure to validate syntax before actually moving to the analysis and code implementation to provide a clearer implementation of functions and code structure. Through research, it is stated that It is both used in computer programming and linguistics to

break down expressions into basic components. **(Opinaldo, 2021)**

REFERENCES:

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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